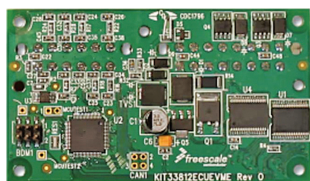


Link. <https://www.electronicsforu.com/electronics-projects/automotive-electronic-control-unit-reference-design>

OEM Brand. Microchip

Small engine control design

This ECU reference design enables you to develop and test engine management systems for motorcycles, scooters, generators, pumps, and other small-engine applications. The platform enables rapid prototyping of electronic fuel injection



systems, allowing you to create fuel maps, optimise injection timing, and evaluate engine performance before designing custom hardware. It can also be used to develop ignition control algorithms, test spark timing strategies, and validate engine operation under different conditions. By supporting inputs from position, temperature, pressure, and throttle sensors, the design helps you build and verify real-time control algorithms for improved fuel efficiency and engine performance. The integrated diagnostic and fault-monitoring features enable the development of maintenance, fault logging, and service functions, while the communication interface supports ECU calibration and diagnostic tool development.

Applications. You can use this design to study proven power-supply, protection, and control architectures, thereby reducing development time and accelerating the creation of production-ready ECU solutions.

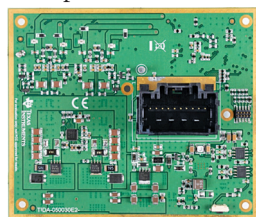
Link. <https://www.electronicsforu.com/electronics-projects/small-engine-control-reference-design>

OEM Brand. NXP Semiconductor

Automotive headlight ECU design

This reference design enables you to develop automotive headlight and lighting control units that support

matrix LED, adaptive beam, and rear-light applications. Using a high-efficiency two-stage power architecture, you can drive multiple LED channels with pixel-level control, create



dynamic lighting patterns, manage high-power LED loads, and implement intelligent lighting functions. The design also supports vehicle network communication, allowing seamless integration with other automotive systems. You can use it to evaluate power management, thermal performance, electromagnetic compatibility, and overall system efficiency, helping accelerate the development of advanced automotive lighting ECUs.

Applications. The reference design finds applications in automotive lighting, including dynamic and static headlights and rear lights.

Link. <https://www.electronicsforu.com/electronics-projects/automotive-headlight-ecu-reference-design>

OEM Brand. Texas Instruments (TI)

Vehicle control unit design

The vehicle control unit reference design provides automotive engineers with a ready-made platform for building, testing, and validating



vehicle control and communication systems without starting

from scratch. You can use it to connect and manage various vehicle components, such as ECUs, sensors, and control modules, and to study how data flows within a vehicle. It helps you design and test central controllers and gateways that manage communication between many systems, and compare different vehicle network styles, such

as centralised and zonal setups. You can also connect various sensors and devices, write and test software drivers, and verify how the system behaves under different conditions. It supports testing of common vehicle communication links, allowing you to assess how well different parts share data and work together across systems such as engine, body, chassis, and safety. The platform also allows testing of fast in-vehicle networks, helping you build systems for data sharing, updates, and communication between main computers and other vehicle units. You can link it with other automotive boards to create realistic vehicle setups and test how hardware and software work together before final products are developed.

Applications. This reference design can be used in ECU communication testing, sensor integration and validation, in-vehicle networking, gateway and zonal architecture development, and ADAS and safety system development.

Link. <https://www.electronicsforu.com/electronics-projects/vehicle-control-unit-reference-design>

OEM Brand. Renesas

These reference designs provide automotive design engineers with a practical starting point for building and testing various ECU functions without having to start from scratch. Instead of spending time on basic architecture, you can focus on improving performance, safety, communication, and system integration across power management, control, lighting, security, and networking domains. They also make it easier to validate hardware and software together under real-world conditions, helping reduce development time, minimise risk, and accelerate product development. **EFY**

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